

Problem 2

Part a)

We want to show that $\varphi_X(2u) = \varphi_X(u)^3 \varphi_X(-u)$ for all $u \in \mathbb{R}$. By Theorem 3.5.2b, we have $\varphi_{2X}(u) = \varphi_X(2u)$. We observe that $2X = (X + Y) + (X - Y)$. Since $X + Y$ and $X - Y$ are independent, we can apply Lemma 3.5.6 directly to the characteristic function of their sum:

$$\begin{aligned}\varphi_X(2u) &= \varphi_{(X+Y)+(X-Y)}(u) \\ &= \varphi_{X+Y}(u) \cdot \varphi_{X-Y}(u)\end{aligned}\tag{1}$$

Next, we evaluate $\varphi_{X+Y}(u)$ and $\varphi_{X-Y}(u)$ separately:

$$\begin{aligned}\varphi_{X+Y}(u) &= \varphi_X(u) \cdot \varphi_Y(u) && \text{(by Lemma 3.5.6, independence of } X \text{ and } Y) \\ &= \varphi_X(u)^2 && \text{(since } X \text{ and } Y \text{ are identically distributed)}\end{aligned}$$

$$\begin{aligned}\varphi_{X-Y}(u) &= \varphi_{X+(-Y)}(u) \\ &= \varphi_X(u) \cdot \varphi_{-Y}(u) && \text{(by Lemma 3.5.6, independence of } X \text{ and } -Y) \\ &= \varphi_X(u) \cdot \varphi_Y(-u) && \text{(by Theorem 3.5.2b)} \\ &= \varphi_X(u) \cdot \varphi_X(-u) && \text{(since } X \text{ and } Y \text{ are identically distributed)}\end{aligned}$$

Combining these results back into (1) gives:

$$\begin{aligned}\varphi_X(2u) &= \varphi_{X+Y}(u) \cdot \varphi_{X-Y}(u) \\ &= \varphi_X(u)^2 \cdot \varphi_X(u) \cdot \varphi_X(-u) \\ &= \varphi_X(u)^3 \cdot \varphi_X(-u)\end{aligned}$$

□

Part b)

Step 0: Show that $\psi(u) := \frac{\varphi_X(u)}{\varphi_X(-u)}$ is well-defined. We are given the hint $\psi(u) := \frac{\varphi_X(u)}{\varphi_X(-u)}$, but we must first ensure $\varphi_X(u) \neq 0$ for all $u \in \mathbb{R}$. Setting $a := \mathbb{E}(\cos(uX))$ and $b := \mathbb{E}(\sin(uX))$, we have $\varphi_X(u) = a + bi$ and $\varphi_X(-u) = a - bi$ (by Definition 3.5.1), so:

$$\varphi_X(u) \cdot \varphi_X(-u) = (a + bi)(a - bi) = a^2 + b^2 = |\varphi_X(u)|^2$$

Applying part a) at u and $-u$ gives:

$$\begin{aligned}|\varphi_X(2u)|^2 &= \varphi_X(2u)\varphi_X(-2u) \\ &= [\varphi_X(u)^3 \varphi_X(-u)] [\varphi_X(-u)^3 \varphi_X(u)] \\ &= \varphi_X(u)^4 \cdot \varphi_X(-u)^4 \\ &= (\varphi_X(u) \cdot \varphi_X(-u))^4 \\ &= (|\varphi_X(u)|^2)^4 \\ &= |\varphi_X(u)|^8.\end{aligned}$$

Hence $|\varphi_X(2u)| = |\varphi_X(u)|^4$. If $\varphi_X(u_0) = 0$ for some $u_0 \in \mathbb{R}$, then substituting $u = u_0/2$ into $|\varphi_X(2u)| = |\varphi_X(u)|^4$ gives $|\varphi_X(u_0)| = |\varphi_X(u_0/2)|^4 = 0$, so $\varphi_X(u_0/2) = 0$. By induction $\varphi_X(u_0/2^n) = 0$ for all $n \in \mathbb{N}$. But $u_0/2^n \rightarrow 0$ and, by Theorem 3.5.2a, $\varphi_X(0) = 1$, contradicting continuity (Theorem 3.5.2c). Therefore $\varphi_X(u) \neq 0$ for all $u \in \mathbb{R}$, and ψ is well-defined.

Step 1: Show that $\psi(2u) = \psi(u)^2$ for all $u \in \mathbb{R}$. Replacing u with $2u$ in the definition of ψ gives:

$$\psi(2u) = \frac{\varphi_X(2u)}{\varphi_X(-2u)} \quad (2)$$

From Part a), we have:

$$\varphi_X(2u) = \varphi_X(u)^3 \varphi_X(-u) \quad (3)$$

Substituting u with $-u$ into (3) yields:

$$\varphi_X(-2u) = \varphi_X(-u)^3 \varphi_X(u) \quad (4)$$

Replacing (3) and (4) back into (2) gives:

$$\psi(2u) = \frac{\varphi_X(2u)}{\varphi_X(-2u)} = \frac{\varphi_X(u)^3 \varphi_X(-u)}{\varphi_X(-u)^3 \varphi_X(u)} = \frac{\varphi_X(u)^2}{\varphi_X(-u)^2} = \psi(u)^2 \quad (5)$$

Step 2: Using induction, show that $\psi(u) = \psi(u/2^n)^{2^n}$ for all $n \in \mathbb{N}$. For the base case $n = 1$, we replace u with $\frac{u}{2}$ in (5), to obtain:

$$\psi(u) = \psi(u/2)^2 \quad (6)$$

Assume $\psi(u) = \psi(u/2^n)^{2^n}$ holds for some $n \in \mathbb{N}$. Then:

$$\begin{aligned} \psi(u) &= \psi(u/2^n)^{2^n} \\ \implies \psi(u/2) &= \psi(u/2^{n+1})^{2^n} \\ \implies \psi(u) &= (\psi(u/2^{n+1})^{2^n})^2 && \text{(by (6))} \\ \implies \psi(u) &= \psi(u/2^{n+1})^{2^{n+1}} \end{aligned}$$

Therefore, by induction:

$$\psi(u) = \psi(u/2^n)^{2^n} \quad \text{for all } n \in \mathbb{N} \quad (7)$$

Step 3: Show that $\psi(v) = 1 + \frac{v^2 \delta(v)}{\varphi_X(-v)}$ for $v \in \mathbb{R}$, where δ is some function with $\lim_{v \rightarrow 0} \delta(v) = 0$. Since $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, we can apply the expansion formula of characteristic functions from Theorem 3.5.2d with $p = 2$ and $v \in \mathbb{R}$:

$$\begin{aligned} \varphi_X(v) &= \sum_{k=0}^2 \frac{(iv)^k}{k!} \cdot \mathbb{E}(X^k) + \frac{(iv)^2}{2!} \cdot \epsilon_2(v) \\ &= 1 + iv\mathbb{E}(X) + \frac{(iv)^2}{2} \mathbb{E}(X^2) + \frac{(iv)^2}{2} \epsilon_2(v) \\ &= 1 - \frac{v^2}{2} - \frac{v^2}{2} \epsilon_2(v) \end{aligned} \quad (8)$$

where $\lim_{v \rightarrow 0} \epsilon_2(v) = 0$. Replacing v with $-v$ gives:

$$\varphi_X(-v) = 1 - \frac{v^2}{2} - \frac{v^2}{2} \epsilon_2(-v)$$

Taking the difference gives:

$$\begin{aligned} \varphi_X(v) - \varphi_X(-v) &= -\frac{v^2}{2} (\epsilon_2(v) - \epsilon_2(-v)) = v^2 \delta(v) \\ \implies \varphi_X(v) &= \varphi_X(-v) + v^2 \delta(v) \end{aligned}$$

where $\delta(v) := -\frac{1}{2}(\epsilon_2(v) - \epsilon_2(-v))$ and $\lim_{v \rightarrow 0} \delta(v) = 0$. Thus, we can express $\psi(v)$ as:

$$\begin{aligned}\psi(v) &= \frac{\varphi_X(v)}{\varphi_X(-v)} \\ &= \frac{\varphi_X(-v) + v^2\delta(v)}{\varphi_X(-v)} \\ &= 1 + \frac{v^2\delta(v)}{\varphi_X(-v)}\end{aligned}\tag{9}$$

Step 4: Show that $\psi(u) = 1$ for all $u \in \mathbb{R}$, and deduce that φ_X is symmetric.

Let $n \in \mathbb{N}$ and $m := 2^n$. By the induction result of step 2 (7), we have $\psi(u) = \psi(u/m)^m$. Substituting $v = u/m$ into (9) gives:

$$\begin{aligned}\psi(u) &= \left(1 + \frac{(u/m)^2\delta(u/m)}{\varphi_X(-u/m)}\right)^m \\ &= \left(1 + \frac{1}{m} \left[\frac{(u^2/m)\delta(u/m)}{\varphi_X(-u/m)}\right]\right)^m\end{aligned}\tag{10}$$

Let $x_m := \frac{(u^2/m)\delta(u/m)}{\varphi_X(-u/m)}$. We now analyze the limit of x_m as $m \rightarrow \infty$ (which implies $u/m \rightarrow 0$):

- **Numerator:** As $m \rightarrow \infty$, the term $u^2/m \rightarrow 0$. Furthermore, since $\lim_{v \rightarrow 0} \delta(v) = 0$, we have $\delta(u/m) \rightarrow 0$. Thus, the entire numerator approaches $0 \cdot 0 = 0$.
- **Denominator:** By Theorem 3.5.2c, the characteristic function φ_X is uniformly continuous on $\mathbb{R}$. Because it is continuous at 0, and $\varphi_X(0) = 1$ (Theorem 3.5.2a), we have $\lim_{m \rightarrow \infty} \varphi_X(-u/m) = \varphi_X(0) = 1$.

Because the denominator approaches 1 (and since x_m is well-defined for all m , Step 0), we have that:

$$\lim_{m \rightarrow \infty} x_m = \frac{0}{1} = 0$$

Using the limit theorem for the exponential function ($\lim_{m \rightarrow \infty} (1 + \frac{x_m}{m})^m = e^L$ whenever $\lim_{m \rightarrow \infty} x_m = L$ exists, which remains valid for complex x_m and along the subsequence $m = 2^n$), we take the limit of (10):

$$\lim_{m \rightarrow \infty} \psi(u) = \lim_{m \rightarrow \infty} \left(1 + \frac{x_m}{m}\right)^m = e^0 = 1$$

However, by the definition of $\psi(u) := \frac{\varphi_X(u)}{\varphi_X(-u)}$, the value $\psi(u)$ does not depend on m . Since $\psi(u)$ is a constant with respect to m , its limit as $m \rightarrow \infty$ must simply be its value:

$$\psi(u) = \lim_{m \rightarrow \infty} \psi(u) = 1$$

Also, by the definition of $\psi(u) := \frac{\varphi_X(u)}{\varphi_X(-u)}$, the condition $\psi(u) = 1$ implies $\varphi_X(u) = \varphi_X(-u)$ for all $u \in \mathbb{R}$, i.e., the characteristic function φ_X is symmetric.

Step 5: Show that $\varphi_X(u) = \varphi_X(u/2^n)^{4^n}$ for all $n \in \mathbb{N}$. From Step 4, we know that φ_X is symmetric, meaning $\varphi_X(-u) = \varphi_X(u)$ for all $u \in \mathbb{R}$. We substitute this into the result from Part a):

$$\begin{aligned}\varphi_X(2u) &= \varphi_X(u)^3 \varphi_X(-u) \\ &= \varphi_X(u)^3 \varphi_X(u) \\ &= \varphi_X(u)^4\end{aligned}\tag{11}$$

We now use mathematical induction to show that $\varphi_X(u) = \varphi_X(u/2^n)^{4^n}$ for all $n \in \mathbb{N}$.

Base case ($n = 1$): Replacing u with $u/2$ in (11) yields:

$$\varphi_X(u) = \varphi_X(u/2)^4 \quad (12)$$

which satisfies the claim for $n = 1$.

Inductive step: Assume that $\varphi_X(u) = \varphi_X(u/2^n)^{4^n}$ holds for some $n \in \mathbb{N}$. We must show it holds for $n + 1$. By applying (12) to the argument $u/2^n$, we obtain:

$$\varphi_X(u/2^n) = \varphi_X\left(\frac{u/2^n}{2}\right)^4 = \varphi_X(u/2^{n+1})^4$$

Substituting this back into our inductive hypothesis gives:

$$\begin{aligned} \varphi_X(u) &= (\varphi_X(u/2^{n+1})^4)^{4^n} \\ &= \varphi_X(u/2^{n+1})^{4 \cdot 4^n} \\ &= \varphi_X(u/2^{n+1})^{4^{n+1}} \end{aligned}$$

Therefore, by induction, $\varphi_X(u) = \varphi_X(u/2^n)^{4^n}$ holds for all $n \in \mathbb{N}$.

Step 6: Show that $\varphi_X(u) = e^{-u^2/2}$ and conclude $X \sim \mathcal{N}(0, 1)$. Recall the equation (8) from Step 3 and let $v = u/2^n$ with $n \in \mathbb{N}$:

$$\begin{aligned} \varphi_X(v) &= 1 - \frac{v^2}{2} - \frac{v^2}{2} \epsilon_2(v) \\ \implies \varphi_X(u/2^n) &= 1 - \frac{(u/2^n)^2}{2} - \frac{(u/2^n)^2}{2} \epsilon_2(u/2^n) \end{aligned} \quad (13)$$

where $\lim_{n \rightarrow \infty} \epsilon_2(u/2^n) = 0$. Substituting (13) into the expression $\varphi_X(u) = \varphi_X(u/2^n)^{4^n}$ from Step 5 gives:

$$\varphi_X(u) = \left(1 - \frac{(u/2^n)^2}{2} - \frac{(u/2^n)^2}{2} \epsilon_2(u/2^n)\right)^{4^n}$$

Let $m = 4^n$ and noting that $(u/2^n)^2 = u^2/4^n = u^2/m$ and $2^n = \sqrt{m}$, we obtain:

$$\begin{aligned} \varphi_X(u) &= \left(1 - \frac{u^2/m}{2} - \frac{u^2/m}{2} \epsilon_2\left(\frac{u}{\sqrt{m}}\right)\right)^m \\ &= \left(1 + \frac{1}{m} \left[-\frac{u^2}{2} - \frac{u^2}{2} \epsilon_2\left(\frac{u}{\sqrt{m}}\right)\right]\right)^m \end{aligned} \quad (14)$$

Let $x_m := -\frac{u^2}{2} - \frac{u^2}{2} \epsilon_2(u/\sqrt{m})$. As $m \rightarrow \infty$, the error term $\epsilon_2(u/\sqrt{m})$ approaches 0. Thus, the limit of x_m is:

$$\lim_{m \rightarrow \infty} x_m = -\frac{u^2}{2} - \frac{u^2}{2}(0) = -\frac{u^2}{2}$$

Because $\varphi_X(u)$ does not depend on m , we can take the limit of (14) as $m \rightarrow \infty$. As in Step 4, we use again the limit theorem of the exponential function to obtain:

$$\varphi_X(u) = \lim_{m \rightarrow \infty} \left(1 + \frac{x_m}{m}\right)^m = e^{-u^2/2}$$

This resulting function, $\varphi_X(u) = e^{-u^2/2}$, is exactly the characteristic function of the standard normal distribution $\mathcal{N}(0, 1)$ (as shown in Example 3.5.3, point 4). Since probability measures are uniquely determined by their characteristic functions (Theorem 3.5.4), then $X \sim \mathcal{N}(0, 1)$. $\square$